COMPLETE SYSTEM ARCHITECTURE & BLUEPRINT

ShopSmart AI Master System Specification

Complete End-to-End Architectural Reference Manual, User Flow & Quota Engine, Hardware Fingerprint Sybil Shields, 8-Tab Admin Dashboard Systems, and Failure Fallbacks.

FRAMEWORK & ENGINE
Next.js 16 (App Router) + Turbopack

SEARCH QUOTA ENGINE
10 Daily Free Pool + 0ms Cache Free

DEVICE FINGERPRINTING
Canvas 2D + WebGL GPU + Audio SHA-256

VIRTUAL TRY-ON (VTON)
3 Free Daily Slots + Modal A100 GPU

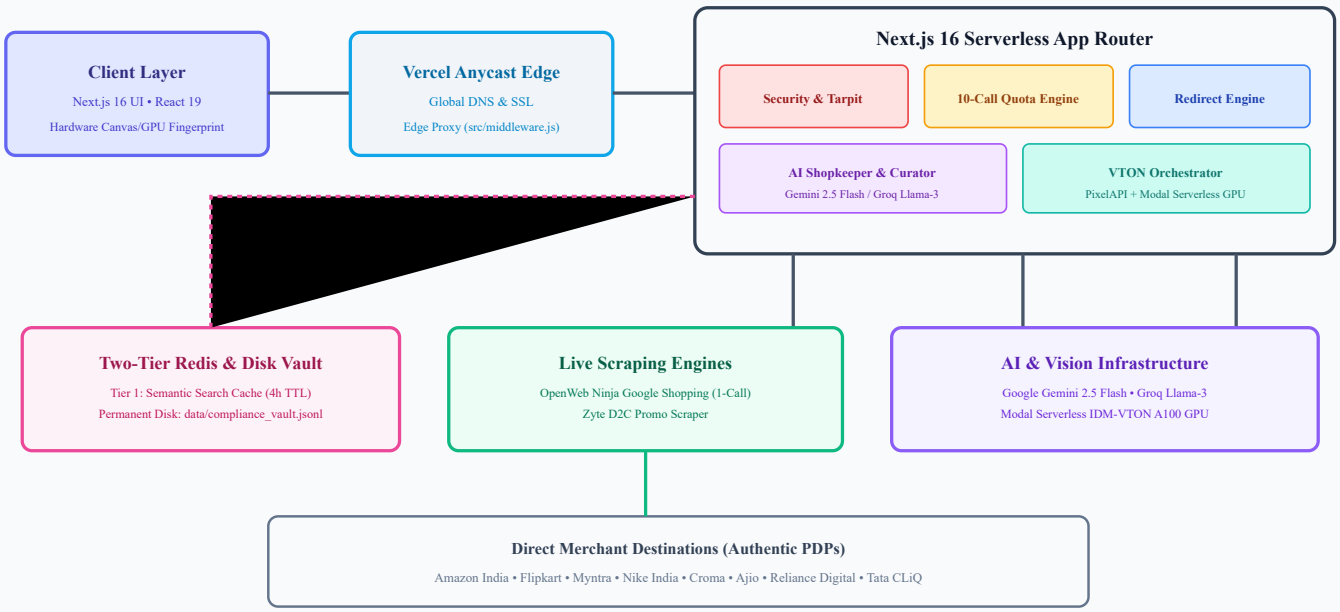
ADMIN COMMAND CENTER
8 Deep Tabs (Telemetry, Crashes, Tarpit)

SECURITY STANDARD
Stealth Tarpit & Permanent Disk Ledger

1. Global High-Level System Topology

ShopSmart AI is engineered as a low-latency, distributed microservice aggregator. It merges client-side hardware fingerprinting, multi-lingual normalization, serverless edge routing, dual-tier Redis caching, and autonomous cloud AI models.

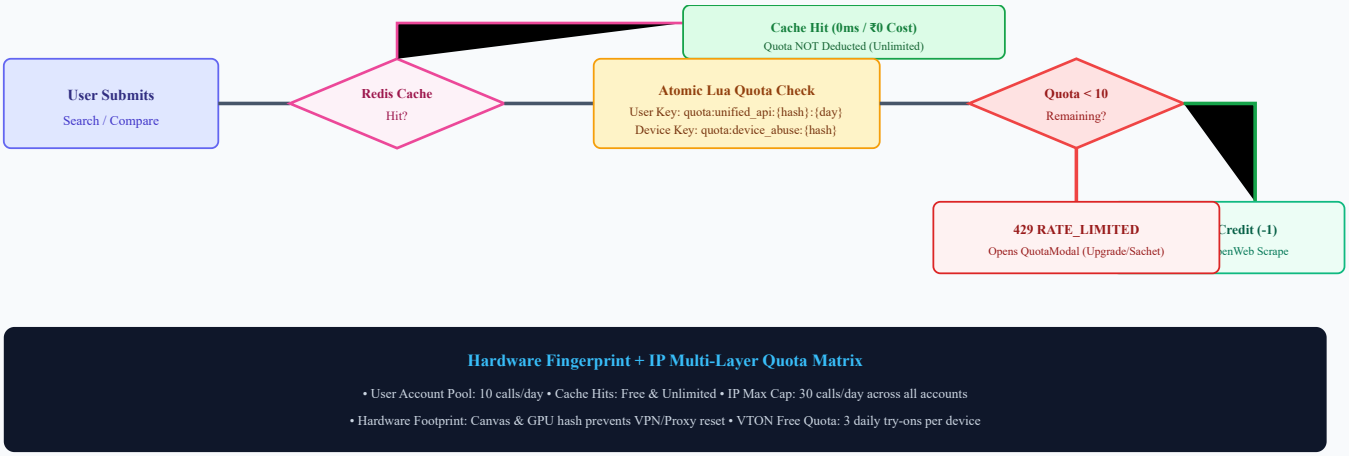
FIGURE 1.1: COMPLETE END-TO-END PLATFORM TOPOLOGY — GLOBAL DISTRIBUTED STACK



2. User Flow & Multi-Tier Quota Architecture

To deliver high quality while preventing API cost overruns and Sybil bot attacks, ShopSmart AI enforces a rigorous multi-tier quota architecture driven by atomic Redis Lua scripts.

FIGURE 2.1: UNIFIED 10-CALL QUOTA & 0MS CACHE BYPASS FLOWCHART — RATE LIMITER ENGINE



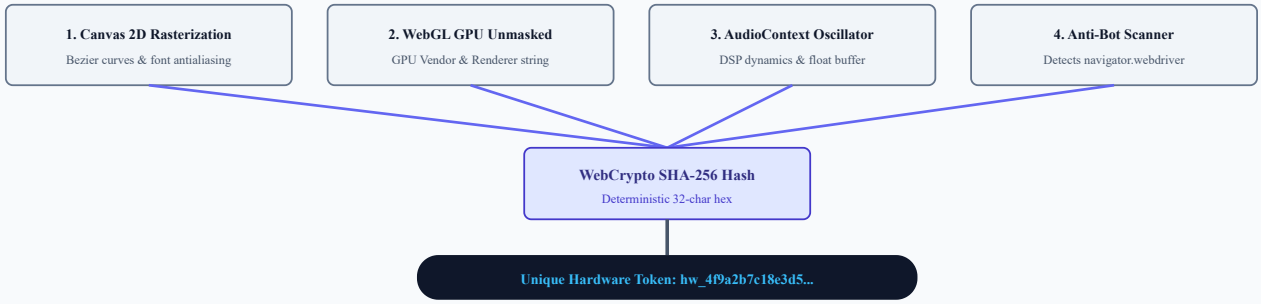
2.1 Complete Quota & Free Tier Rules Table

Tier / User Type	Live Search Quota	Cached Search Quota	Virtual Try-On (VTON)	Abuse Ceiling
Anonymous Visitor	10 calls / IST Calendar Day	**Unlimited (0ms)**	3 free try-ons lifetime / device	30 calls / IP / day ceiling
Clerk Authenticated	10 calls / IST Calendar Day	**Unlimited (0ms)**	3 free try-ons daily (IST Reset)	Hardware Canvas hash locked
Sachet Pack (₹10)	+25 Extra Real-Time Searches	**Unlimited (0ms)**	+5 High-Res AI Try-Ons	Direct Clerk Stripe/Razorpay hook
Admin (user_3JGg...)	**Unlimited**	**Unlimited**	**Unlimited**	Bypasses all tripwires & limits

3. Hardware Fingerprinting & Sybil Defense Engine

To prevent malicious users from bypassing search limits and VTON trials by switching to Incognito mode or rotating VPN IP addresses, ShopSmart AI deploys an invisible client-side Hardware Footprint Engine (`src/lib/deviceFingerprint.js`).

FIGURE 3.1: HARDWARE & SUB-PIXEL CANVAS FINGERPRINT PIPELINE **ENGINE: SRC/LIB/DEVICEFINGERPRINT.JS**



🎨 Sub-Pixel Canvas 2D Entropy

Different operating systems, graphic cards, and font rasterizers render emojis (🐶🛋️) and bezier curves with minuscule sub-pixel differences. This creates a distinct visual entropy without recording any PII.

🎮 WebGL GPU Hardware Extraction

Extracts the unmasked graphic card vendor (e.g., `ANGLE (Apple, Apple M2 Max, OpenGL 4.1)` or `NVIDIA GeForce RTX 4080`). Virtual machines and automated headless bots typically expose `SwiftShader` or `Mesa Offscreen` .

🔊 AudioContext Oscillator Synthesis

Synthesizes an inaudible audio buffer with a DynamicsCompressorNode. The float32 audio output varies across hardware soundcards and mathematical floating-point implementations.

🛡️ 100% Privacy & DPDP 2023 Compliant

No cookies, MAC addresses, or personal names are ever gathered. The hash is completely one-way (SHA-256) and cannot be reversed into user identity.

4. Admin Command Center Deep Specification

The Admin Dashboard (`src/app/admin/page.js`) is a protected, enterprise-grade command portal accessible strictly to verified administrators (`ADMIN_USER_IDS=user_3JGg_MnXc5O3td`). It provides eight deep subsystems:

FIGURE 4.1: 8-TAB ADMIN COMMAND PORTAL LAYOUT **ROUTE: /ADMIN**



4.1 Detailed Admin Capabilities by Tab

Tab Name	What the Operator Can See	Available Actions	Underlying File & API
Analytics & Metrics	Total queries today, cache hit %, average query latency, API cost counter ($), live query stream.	Toggle 5s Auto-Refresh, inspect raw search queries and timestamps.	`src/services/adminMetrics.js` `GET /api/admin/metrics`
Kill-Switches	Live on/off status of VTON Trial Room, AI Concierge, Trust Sheet, Zyte, and Head-to-Head Compare.	Instant toggle on/off (saves to Redis in `admin:settings:feature_flags`).	`src/services/featureFlags.js` `POST /api/admin/features`
Crash Vault	Full error messages, sanitized stack traces, source file line numbers, and error severity.	Mark crash as Resolved, view error distribution across routes.	`src/services/crashVault.js` `GET /api/admin/crashes`
Helpdesk Tickets	Customer reports (Broken Link, Incorrect Price, Feature Request), user email, submit timestamp.	Filter by status (open, in_progress, resolved), write admin response notes.	`src/services/supportTickets.js` `GET /api/support/tickets`
Coupons & Brands	Zyte single-store coupon scraper tester, personal affiliate tags table, aggregator tokens.	Live test scrapers against D2C stores (e.g. `boat-lifestyle.com`), add custom coupons.	`src/services/zyteService.js` `POST /api/admin/settings`
Security Radar	Live threat incidents list, count of throttled IPs, count of monitored IPs, banned set.	1-Click Manual IP Ban (`SADD security:banned_ips`), 1-Click Throttle Clear.	`src/services/securityRadar.js` `GET /api/admin/security`
Link Sentinel	Sentinel probe health, circuit breaker state (CLOSED, OPEN, HALF-OPEN), probe history.	1-Click "Run Sentinel Probe Now" button to manually test & reset circuits.	`src/services/autoHealingSentinel.js` `POST /api/admin/sentinel`
Cache Manager	Total cached search keys, total compare keys, memory footprint estimates.	1-Click Flush Search Cache, 1-Click Flush Compare Cache, Flush Threat Vault.	`src/services/admin.js` `POST /api/admin/settings`